Projects in Life Science Investing

Raymond Chiasson

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Contact

Email	rchiasson@live.ca
Phone	(519) 722-3564
LinkedIn	linkedin.com/raymondchiasson
Website	raymondchiasson.com
GitHub	github.com/rc416
Academic CV	raymondchiasson.com/CV



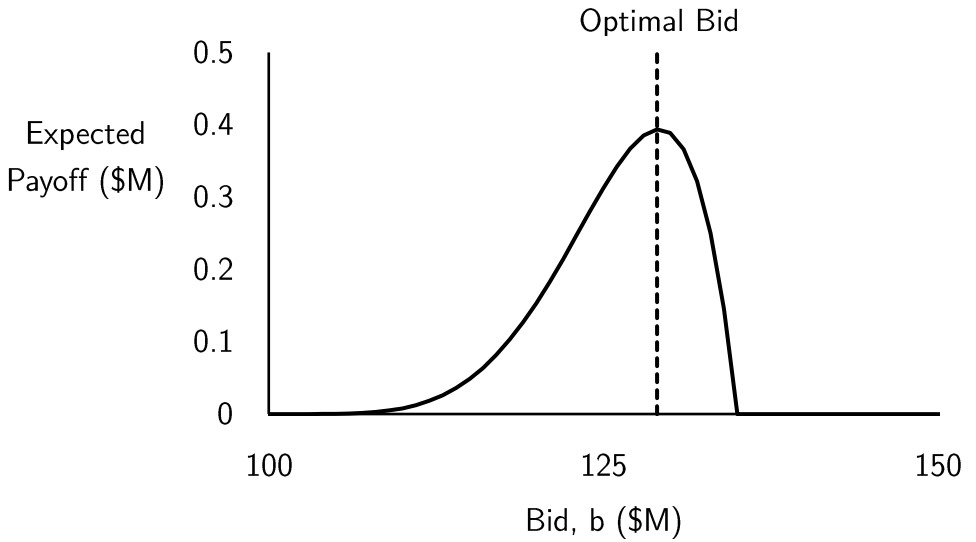


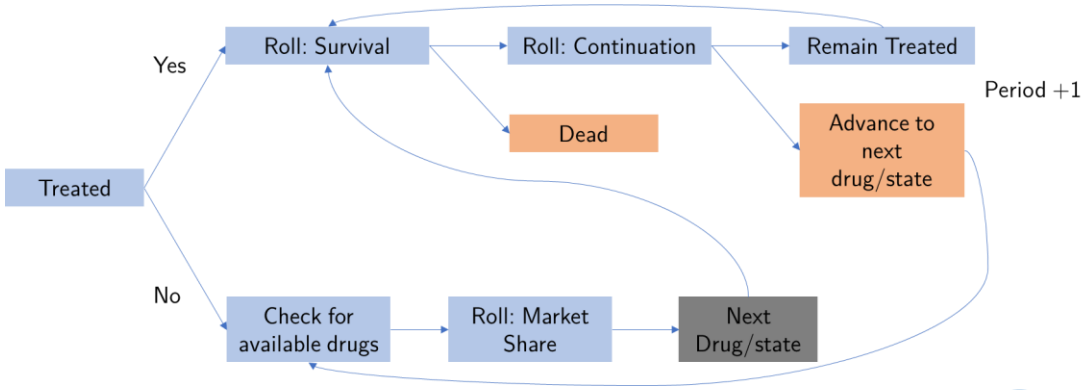




Summary of Key Projects

Project	Summary
1. Optimal bidding strategy for competitive auctions	Mathematically-informed method to systematically incorporate past information and select the optimal bid in an auction.
2. Machine learning to identify investment targets	Automate the process of identifying and validating investment targets using standard machine learning tools.
3. Novel investment strategy	Preliminary financial modeling and market sizing for alternative investment strategy.
4. Dynamic patient model	Template code to implement patient models for forecasting drug sales in rapidly progressing diseases.
5. Training in statistics and coding	Training targeted for analysts to help identify and execute similar projects.





1. Optimal bidding strategy in competitive auctions

Motivation

- Company needed to choose a bid in competitive sealed first price auctions for royalty entitlements.
- Had a sophisticated forecasting operation, but did not have structured way to select a bid.
- Did not have a systematic way to incorporate information from their extensive database of historical transactions.

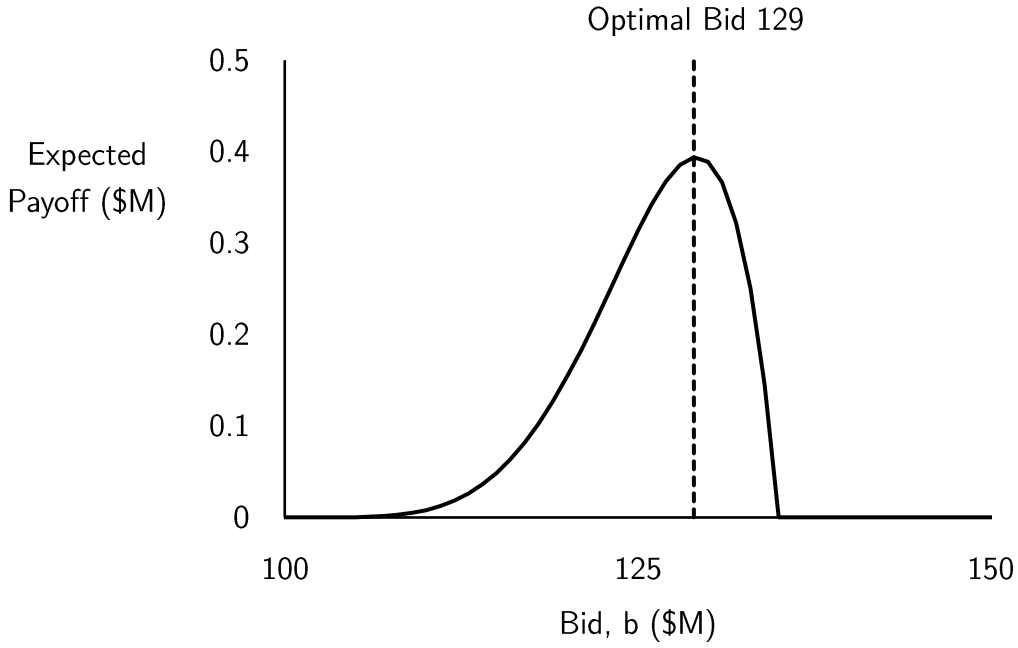
Approach

- Developed systematic approach to select optimal bid which captures the key tradeoff between payoff and probability of winning; i.e. a low bid is more profitable but less likely to win.
- The key challenge is to approximate the probability of winning function using historical data.

Results

- Method to estimate the optimal bid (maximize expected payoff)
- Useful offshoot: establish minimum viable deal size.

$$\begin{aligned} Expected\ Payoff &= Payoff \times Probability\ of\ winning \\ &= (Valuation - bid) \times P(bid) \end{aligned}$$



2. Machine learning to identify investment targets

Motivation

- Company maintained list of 400 drugs that are most likely to be suitable investment targets. This important list dictated which transactions were pursed.
- Considerable manual effort was required to evaluate new and unreviewed drugs for inclusion on this list. Any approved drugs or those in late-stage clinical development needed to be evaluated.

Approach

- Identified list of 700 drugs that had been manually evaluated by the company's current process.
- Combine with list of drug characteristics from commercial databases.
- Trained 4 standard AI algorithms and applied to 3500 untested drugs in the commercial databases.

Results

- High level of accuracy in the training set (>90%).
- Identified list of top investment targets to enrich manual approach.
- Pending validation, approach to replace parts of the manual process.

Drug	Indication	Stage of development	Marketer	...	Label
Keytruda	Oncology	Marketed	Merck	...	Good Investment
Zulresso	Depression	Marketed	Sage Therapeutics	...	Bad Investment
...	...	...	...	...	...
New Drug 1	Depression	Phase II	Merck	...	?
New Drug 2	Diabetes	Marketed	BMS	...	?

Features – observable characteristics of drugs

Label – category or classification of interest



AI model tries to find the relationship between features and labels.

3. Novel investment strategy

Motivation

- Company was looking for alternative investment strategies to increase deployment and improve diversification.

Approach

- Identified novel strategy involving milestones on pharmaceutical products.
- Performed initial financial modeling to understand expected returns, volatility, and purchase prices.
- Analyzed recent milestone transactions to estimate the size of the opportunity.

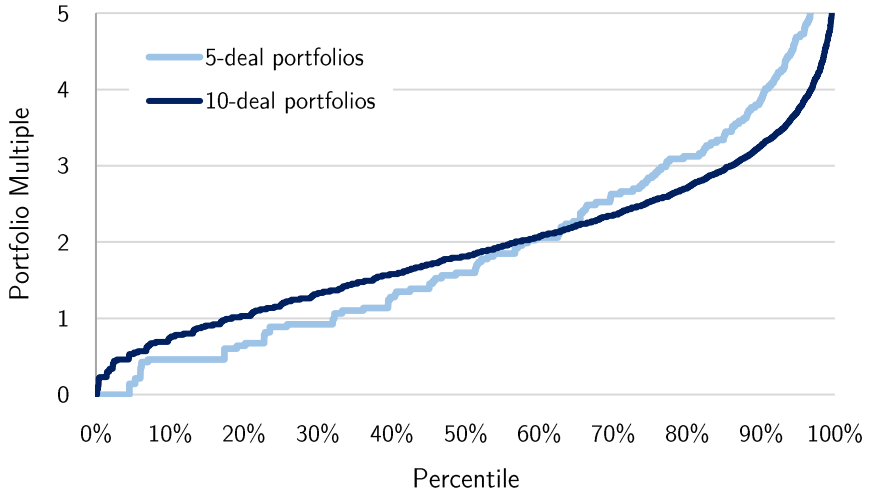
Results

- Modeling and market sizing showed that the opportunity was attractive.
- Preliminary discussions with counterparties to potential transactions is ongoing.

Median portfolio outcomes (45-55th percentiles)

	5-deal	10-deal
Payouts (25 upfront)	39 – 52	48 – 55
Multiple	1.4 – 1.8x	1.7 – 1.9x
IRR	22 – 37%	28 – 36%

Ordered portfolio multiples



4. Dynamic patient model

Motivation

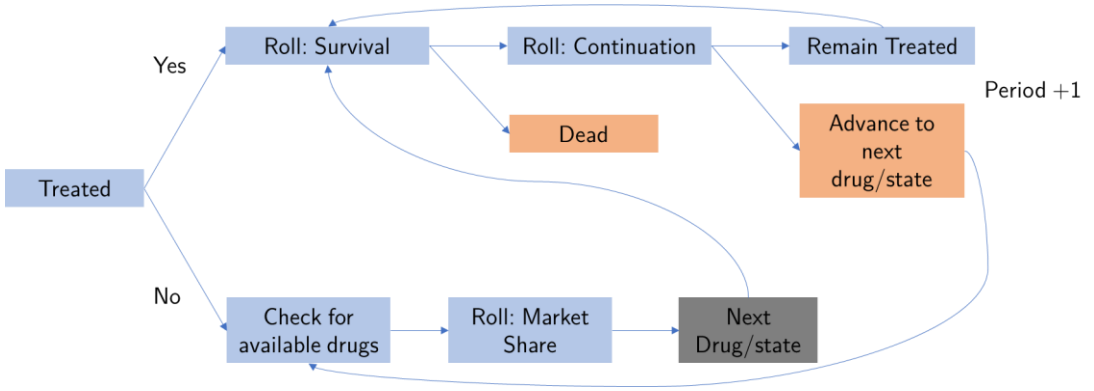
- Company had limited experience implementing dynamic patient models due to personnel departures.
- Lacked the infrastructure for a flexible implementation of dynamic models where forecaster can control the choices and probabilities for drug selection, adherence, and survival.

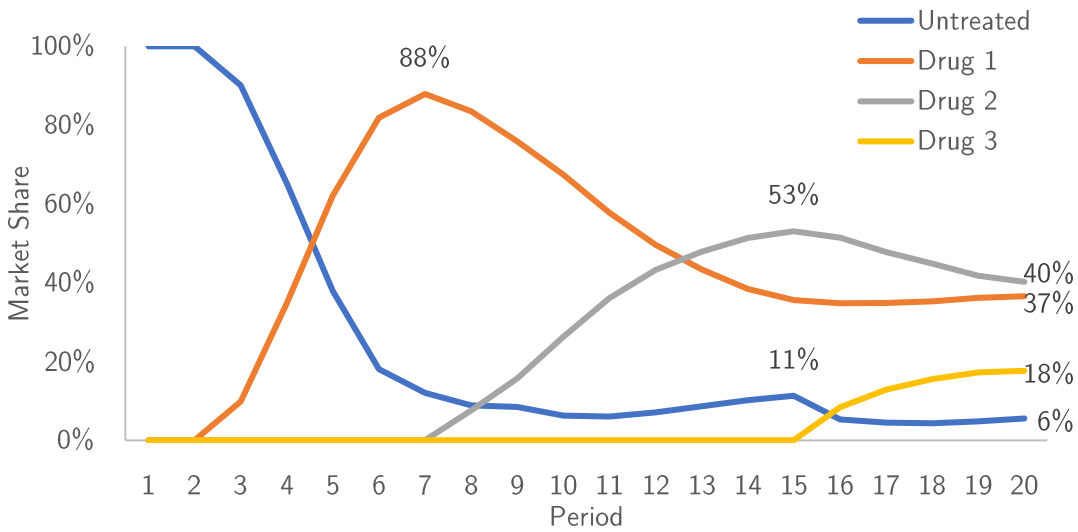
Approach

- Wrote code to implement the desired model with required flexibility.
- Written in high-level language (Python) with object-oriented approach to make the code accessible to beginners.

Results

- Code completed and validated on sample dataset.





5. Training for analysts in statistics and coding

Motivation

- CEO wanted analysts to develop technical skills and find novel projects outside of normal deliverables.
- Analyst team was comprised primarily of those with advanced degrees in life science. Limited training in math, statistics, and programming.
- Lots of "low hanging fruit" for analysts with some basic training.

Approach

- Developed training sessions and reference material.
- Identified 10-20 small projects that were both suitable training exercises and useful to the business.

Results

- Training not yet delivered.

